

# Aswini Devadari

## Data Engineer

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### Summary

Data Engineer with 4+ years of experience, designing scalable **ETL/ELT pipelines**, real-time streaming architectures, and cloud-native data solutions on Azure, with deep expertise in **Databricks, Delta Lake, and Unity Catalog for enterprise data governance. Proficient in Python, PySpark, Apache Spark, Apache Kafka, Airflow, and Azure Data Factory**, AWS with hands-on experience building Medallion Lakehouse architectures on Databricks and Delta Lake. Certified Databricks Associate Data Engineer with a strong foundation in batch and real-time data processing, data quality, and cross-functional stakeholder delivery.

### Education

#### Fairleigh Dickinson University

New Jersey, USA

*Master of Science in Computer Science.*

Relevant Coursework: Python, SQL, Data Mining, Cloud Computing, Big Data Analytics

#### JNTUK

India

*Bachelor of Technology in Computer Science*

### Experience

#### Data Engineer | Quantum | Remote | Aug 2025 – June 2026

- **Analysed 250K+** retail transaction records using Python (Pandas), SQL, and PySpark to identify customer purchasing patterns, improve data quality, and support business decision-making.
- Built **PySpark data transformation pipelines** for cleansing, deduplication, and feature engineering on large retail datasets, reducing manual preprocessing effort and improving processing efficiency.
- Processed structured retail datasets stored on **cloud-based storage (AWS S3/Azure Blob Storage)** and optimized data ingestion workflows for downstream analytics.
- Developed customer segmentation and purchasing behaviour models using **Python and SQL**, enabling targeted marketing strategies and sales trend analysis.
- **Performed data validation, quality checks**, and reconciliation using Python scripts to ensure accurate reporting across analytical datasets.
- Evaluated benchmark store performance using transaction and sales data to support uplift testing initiatives and delivered **KPI-driven** insights through Power BI dashboards.
- Collaborated with cross-functional stakeholders to translate business requirements into scalable data transformation workflows and automated reporting solutions.

#### Research Data Engineer | Fairleigh Dickinson University | New Jersey, USA | Jan 2025 – May 2025

- **Engineered end-to-end ETL pipelines** for healthcare and sales datasets using Python and SQL, processing 200K+ records/batch with automated data validation and quality checks; optimized SQL queries improving execution performance by 35%.
- **Built PySpark transformation workflows** on Databricks for large-scale batch ingestion using Medallion (Bronze– Silver–Gold) architecture, reducing manual effort by 40% and pipeline runtime by 25%.
- **Automated incremental data loads** via Azure Data Factory, eliminating manual intervention and ensuring reliable pipeline scheduling across source systems.
- Implemented **schema validation and schema drift handling** mechanisms for evolving source datasets, ensuring reliable ingestion of structured and semi-structured data across ETL pipelines.
- Managed **Databricks cluster configuration** including autoscaling policies and job scheduling via Databricks Workflows to ensure reliable pipeline execution

#### Healthcare Data Engineer | Kaiser Permanente (via Accenture) | India | Jan 2022 – Jan 2024

- **Developed SQL-based ETL workflows** ingesting 500K+ healthcare records from multiple upstream systems into a centralized data warehouse; automated pipelines cutting manual reporting effort by 30%.
- **Implemented HIPAA-compliant anomaly detection** across finance, clinical, and operations domains, enforcing data quality and compliance standards at scale.
- **Improved query execution by 40%** through index optimization, partition pruning, and SQL query restructuring across high-volume healthcare datasets.

- **Delivered clean, transformed datasets** to Power BI dashboards supporting 5+ internal stakeholder teams across clinical and operations functions.

## Data Engineering Trainee | Accenture | India | Jun 2021 – Dec 2021

- **Built Python and SQL ETL scripts** automating data cleaning workflows across 3+ source systems; resolved 15+ data quality issues and reduced manual effort by 20%.
- Gained foundational experience in pipeline design, data extraction, transformation, and loading into structured targets.

## Projects

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### Healthcare Data Quality & Reporting Pipeline | *Databricks · SQL · Azure Data Factory · Power BI*

- **Developed a healthcare Data Engineering pipeline** using Databricks, Spark, and SQL to process and validate 750K+ patient and operational records, improving reporting accuracy across clinical analytics teams.
- **Automated ETL and incremental data ingestion** workflows through Azure Data Factory and Delta Lake integration, reducing manual reconciliation effort by 35% and improving daily data availability.
- **Implemented data quality validation** and transformation logic for Power BI reporting datasets, decreasing reporting inconsistencies by 30% and accelerating stakeholder dashboard delivery timelines.

### Real-Time Streaming Pipeline – Retail Sales Analytics | *Kafka · Spark Structured Streaming · Airflow · AWS S3 · Python*

- **Engineered a real-time Data Pipeline using Apache Kafka and Spark Structured Streaming** to process 50K+ retail events per hour, enabling near real-time sales and operational analytics.
- **Orchestrated streaming ETL workflows** through Apache Airflow DAGs and automated Parquet data delivery into AWS S3, improving pipeline scheduling reliability and downstream accessibility.
- **Optimized event ingestion and distributed processing** logic using Python and Spark transformations, achieving sub-2-second latency for Business Intelligence reporting workflows.

### Cloud ELT with Medallion Architecture – E-Commerce Orders | *Databricks · PySpark · Delta Lake · AWS S3 · Azure Data Factory*

- **Built scalable ELT workflows on Databricks using PySpark and Delta Lake** to process 1M+ e-commerce transaction records, improving large-scale data integration and analytics readiness across reporting systems.
- **Automated incremental data ingestion and transformation** processes through Azure Data Factory and AWS S3 integration, reducing manual intervention and improving data consistency across reporting systems.
- **Tuned Big Data transformation** workloads using broadcast joins, caching, and partition optimization techniques, decreasing end-to-end processing runtime by 45% for analytics consumption.

## Certification

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Databricks Certified Data Engineer Associate

Credential: <https://credentials.databricks.com/0d48ab92-49aa-425f-b5a4-0a2cb0755234#acc.f2Edgn0k>

## Skills

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**Languages & Processing:** Python (Pandas, NumPy), SQL, PySpark, Apache Spark (Core, Spark SQL, Structured Streaming, DataFrame API)

**Cloud Platforms:** Azure (Data Lake Storage Gen2, Synapse Analytics, Data Factory, Stream Analytics, Blob Storage), AWS (S3, EC2, Glue, Redshift, Lambda, Kinesis, Athena, IAM, CloudWatch, Lake Formation), Microsoft Fabric

**Data Platforms & Lakehouse:** Databricks, Delta Lake, Snowflake, Unity Catalog, Medallion Architecture (Bronze–Silver–Gold), Data Lake, Auto Loader

**Streaming & Big Data:** Apache Kafka, Spark Structured Streaming, Real-Time Data Pipelines, Batch & Streaming Processing

**ETL/ELT & Data Engineering:** Data Ingestion, Transformation, Validation, Data Modeling, Star/Snowflake Schema, Incremental Loads

**Orchestration & BI:** Apache Airflow (DAGs), Azure Data Factory, Power BI

**API & Containerization:** Conceptual understanding of REST API design, containerized deployments (Docker), and Kubernetes

**Data Governance:** Data Quality, HIPAA Compliance, Anomaly Detection, Data Validation

**DevOps & Collaboration:** CI/CD Pipelines, Docker, Git, GitHub, Agile/Scrum, MySQL

**AI-Assisted Development:** Familiar with LLM tools (e.g., Claude, ChatGPT, GitHub Copilot) for code generation, debugging, and productivity support